

PROJECT STRATEGY REPORT

CARS4MARS AFRICAN ROVER CHALLENGE 2026

UCT Space Society

Cape Town, South Africa

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Team Leads

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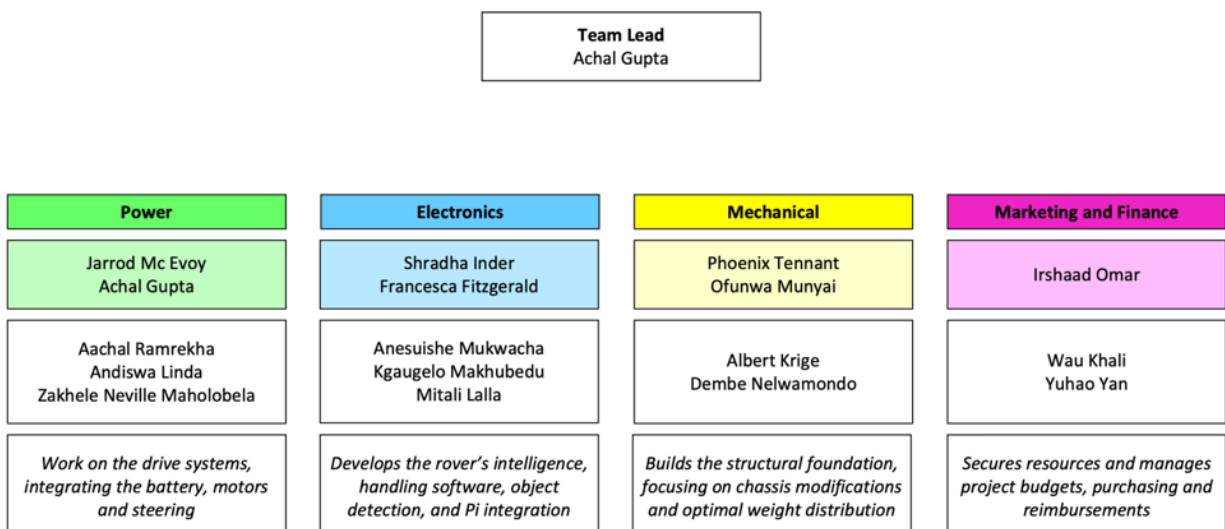
Team Introduction

The UCT Space Society Cars4Mars team is a proudly multidisciplinary student group consisting of seventeen members spanning multiple faculties and degrees at the University of Cape Town (UCT). The team's structure has been purposefully streamlined into four core sub-teams: Power, Electronics, Mechanical, and Marketing & Finance.

The team is led by Achal Gupta, Chairperson of the UCT Space Society, bringing broader leadership experience and vision into the heart of this project. Operating without a formalized Faculty Advisor, the team relies on the collective expertise of its sub-team leaders. No further recruitment is planned at this stage, as the current 17-member roster possesses the exact multifaceted skills necessary for seamless software-hardware coordination.

Our operations are divided across these specialized groups to ensure efficient development. The Power sub-team is responsible for the drive systems, integrating the battery, motors, and steering mechanisms. Our Electronics sub-team develops the rover's intelligence, handling software stack implementation, and real-time object detection. Members within these technical divisions bring diverse, hands-on experience in circuit design, power management, embedded systems, and using computer vision software like OpenCV to build autonomous logical agents.

The Mechanical sub-team builds the structure, focusing on chassis modifications, suspension geometry, and optimal weight distribution. Their members draw on valuable prior mechanical experience and are highly skilled in utilizing CAD software for rapid prototyping. Finally, the Marketing & Finance sub-team secures critical resources, manages project budgets, and oversees purchasing and reimbursements. They are adept at writing, editing, and using design tools like Figma, Canva, and Photoshop for successful outreach and social media management.



Project Management Methodology

To efficiently manage the rover's development alongside our academic commitments, we utilize a structured, part-time student workflow centralized entirely around the Microsoft 365 ecosystem. Learning from our previous project cycle, we have completely overhauled our scheduling and task allocation. By officially restructuring the team into four focused divisions, we ensure that we do not swamp any single member with work, allowing individual technical expertise to shine while carefully accommodating everyone's strict academic commitments. Furthermore, we have adjusted our master schedule to allocate significantly more time to the physical build and integration phases, actively mitigating the rushed timeline that hindered last year's progress.

After evaluating various project management platforms, we purposefully avoided costly subscription-based software in favour of a custom tracking system built in Excel. This database-like spreadsheet serves as our primary tool for managing master project timelines, tracking individual task progress, and monitoring our budget strictly. All documentation, design files, and reports are collaborated on and stored via SharePoint and Microsoft OneDrive, ensuring reliable version control and team-wide visibility across all four sub-teams.

Communication is tailored to be as efficient as possible. We hold weekly full-team meetings hosted on Microsoft Teams to review overall project progress, address critical blockers, and align on upcoming deliverables. For rapid, day-to-day operations, quick decisions, and immediate cross-team troubleshooting, we rely on dedicated WhatsApp groups. This dual-layered communication ensures that formal updates are documented while keeping development agile and responsive.

Task delegation and oversight are driven by our internal leadership structure. Each of our four sub-teams operates semi-independently to execute their specific technical or logistical tasks. The respective sub-team leads act as primary liaisons, communicating directly with the Team Lead to ensure that mechanical, electrical, and software components integrate seamlessly at every major milestone. Furthermore, the Marketing & Finance sub-team is tasked with maintaining the master schedule and flagging any milestone slippage during our weekly syncs. Operating completely independently without a formalized Faculty Advisor, our team relies heavily on internal accountability, rigorous peer review, and the collective experience of our leadership to keep the Cars4Mars project on track.

Resources

The assembly, testing, and development of our rover are distributed across several key facilities at UCT. We have full access to the UCT Electrical Engineering (EEE) laboratories, which provide essential soldering stations, power supplies, and benchtop testing equipment for our electronics integration. Additionally, we have successfully secured permission from the UCT Physics Department to utilize dedicated laboratory space for our main rover assembly and secure component storage. To accommodate the manufacturing of our upgraded structural components, we are also currently attempting to secure access to the main UCT Engineering workshops for heavy metal fabrication and machining.

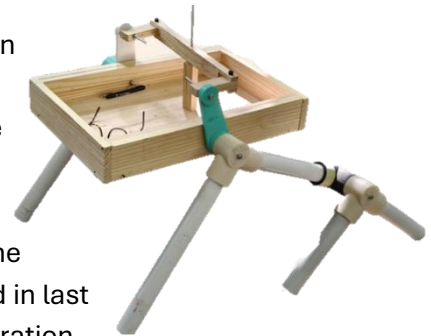
Resource acquisition and budget tracking are strictly overseen by our Marketing & Finance sub-team. This year, the project is strongly supported by a ZAR 15,000 budget allocation that has already been provided by the UCT Space Society, and already utilized to source motors, microcontrollers, electrical components, and prototyping materials. Furthermore, we have received highly valuable in-kind sponsorship, most notably our primary power system (a 12.8V LiFePO₄ battery) which was generously sponsored by the UCT Astronomy Department. However, due to significant early expenditures on essential hardware and components, we are actively looking for additional corporate sponsors and exploring alternative funding sources to sustain our upcoming testing and iteration phases.

For digital resources and engineering design, the team leverages a suite of UCT-licensed programs and accessible tools. We rely on the Microsoft 365 ecosystem for all team-wide collaboration, documentation, and file sharing. This combination of secured physical workspaces, foundational university funding, and robust software gives us the resources necessary to execute our technical goals.

Initial Design Ideas

For the 2026 Cars4Mars competition, our core design strategy is a direct, calculated continuation of our team's work from last year. Rather than discarding past efforts, our design directly builds upon the foundational architecture of our unfinished 2025 Cars4Mars rover prototype—a build we were ultimately unable to compete with. We have spent considerable time stress-testing and analyzing this physical prototype (as seen below) to clearly understand its mechanical shortfalls, allowing us to implement targeted, evidence-based improvements for this year's finalized competition model. The vehicle is engineered to navigate uneven terrain while adhering strictly to the 80 by 80 cm size limit and the 40 kg weight constraint. We spent time building the prototype (as can be seen below) to understand the shortfalls and potential improvements to be made and we arrived at the following conclusions.

Mechanically, we retained last year's 6-wheel rocker-bogie suspension system to maximize traction and stabilize the chassis. The Mechanical sub-team had constructed a PVC prototype to validate the kinematics, leading to plans to upgrade to machined metal components for the final build. The drivetrain utilizes a 4-wheel drive (4WD) configuration, with the front and rear wheels being powered. The electrical system is powered by the same 12.8V LiFePO4 battery used in last year's model, which is already in our possession and undergoing integration.



Our computing stack carries over last year's core hardware: a Raspberry Pi 4B paired with an Arduino Mega. The Arduino manages low-level motor control and sensor polling, while the Raspberry Pi serves as the primary compute node. A Python-based software stack runs on the Raspberry Pi to handle real-time object detection for identifying specific mission targets, such as a hammer, tennis ball, and traffic cone. This is possible due to our use of the Raspberry Pi Camera v3 which provides high-resolution real-time footage and integrates well with our Raspberry Pi out of the box. This stack also manages autonomous navigation through balloon colour detection algorithms. Remote operation and telemetry are handled via a Wi-Fi link to the command station, utilizing a dedicated router provided by the team. In summary the Raspberry Pi serves as the brain of the system and the Arduino serves as the heart of the system.